

Financial Intelligence Platform

Analyst Guide

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Table of Contents

1. Introduction
 2. System Overview
 3. Getting Started
 4. Streamlit Dashboard Navigation
 5. Using the Investment Screener
 6. Generating PDF Tearsheets
 7. Using the REST API
 8. Troubleshooting
 9. Best Practices
 10. Conclusion
-

1. Introduction

1.1 Purpose of this Guide

This Analyst Guide provides comprehensive instructions for using the Financial Intelligence Platform. The platform is designed to simplify financial statement analysis by transforming raw financial data into meaningful insights through automated analytics, interactive dashboards, downloadable reports, and REST APIs.

This guide explains how to operate the major components of the system, including the Streamlit dashboard, investment screener, PDF report generation, and REST API. It also provides troubleshooting guidance for common issues encountered during deployment and usage.

The document is intended for first-time users as well as analysts who wish to explore the analytical capabilities of the platform efficiently.

1.2 Intended Audience

This guide is suitable for:

- Financial Analysts
- Equity Research Analysts
- Investment Professionals
- Students studying Finance or Data Analytics
- Researchers
- Developers integrating with the REST API

No prior knowledge of the project's internal implementation is required to use this guide.

1.3 Platform Features

The Financial Intelligence Platform offers the following capabilities:

- Interactive financial dashboard
- Company-wise financial analysis
- Investment screening using predefined and custom filters
- Peer comparison and benchmarking
- Sector-level financial analysis
- Cash flow intelligence
- Capital allocation analysis
- Automated company tearsheets
- Portfolio summary reports
- REST API for external integrations

These features allow analysts to evaluate companies efficiently without manually processing financial statements.

2. System Overview

The platform follows a structured workflow from data ingestion to report generation.

Raw financial datasets are imported into the system using the ETL pipeline. After validation and cleaning, the data is stored in a SQLite database. The Analytics Engine computes financial ratios, growth metrics, valuation indicators, peer rankings, and cash flow intelligence. The processed information is then made available through the Streamlit dashboard, REST API, and downloadable reports.

The high-level workflow is shown below.

Excel Financial Data



ETL Pipeline



SQLite Database



Analytics Engine



▼ ▼
Dashboard REST API

|

▼

PDF Reports

The user typically interacts with the dashboard while the underlying database and analytics modules perform all computations automatically.

3. Getting Started

3.1 System Requirements

Before using the platform, ensure that the following software is installed.

- Python 3.11 or later
 - Git
 - SQLite
 - pip
 - Streamlit
 - FastAPI
 - Modern web browser
-

3.2 Installing the Project

Clone the repository.

```
git clone <https://github.com/bhimishettihith/financial-intelligence-platform>
```

```
cd financial_intelligence_platform
```

Create a virtual environment.

```
python -m venv venv
```

Activate the virtual environment.

Windows

```
venv\Scripts\activate
```

Linux/macOS

```
source venv/bin/activate
```

Install all required dependencies.

```
pip install -r requirements.txt
```

3.3 Loading Financial Data

Before using the dashboard, financial data must be loaded into the SQLite database.

Run the ETL pipeline.

```
python src/etl/load_database.py
```

The ETL pipeline performs the following operations.

- Reads Excel datasets
- Validates input data
- Removes inconsistencies
- Normalises company information
- Loads validated records into SQLite

After successful execution, the database is ready for analytical processing.

3.4 Generating Financial Analytics

Run the analytics engine.

```
python src/analytics/populate_financial_ratios.py
```

This process computes:

- Financial ratios
- CAGR metrics
- Valuation indicators
- Composite scores
- Cash flow intelligence
- Peer analytics

The generated data becomes available in both the dashboard and REST API.

3.5 Launching the Dashboard

Start the Streamlit application.

```
streamlit run src/dashboard/app.py
```

Once the application starts successfully, it opens automatically in the default web browser.

The dashboard serves as the primary interface for analysing companies.

3.6 Launching the REST API

Start the FastAPI server.

```
uvicorn src.api.main:app --reload
```

After the server starts, the interactive API documentation can be accessed using:

```
http://localhost:8000/docs
```

The Swagger interface allows users to test API endpoints directly from the browser without additional software.

4. Streamlit Dashboard Navigation

The Streamlit dashboard is the primary user interface of the Financial Intelligence Platform. It enables users to analyse companies, compare financial performance, run investment screeners, explore peer groups, and generate analytical reports without interacting directly with the database.

The dashboard is divided into multiple pages, each designed for a specific analytical purpose

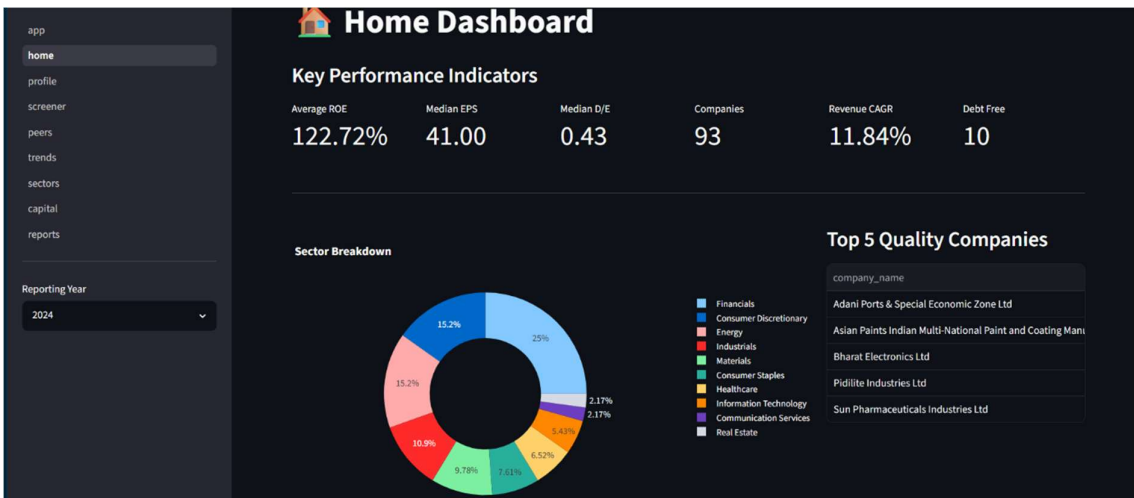
4.1 Home Page

The Home page provides an overview of the Financial Intelligence Platform.

It typically displays:

- Platform introduction
- Total companies available
- Number of sectors
- Available analytical modules
- Navigation links

The Home page serves as the starting point for all analyses.



4.2 Company Profile

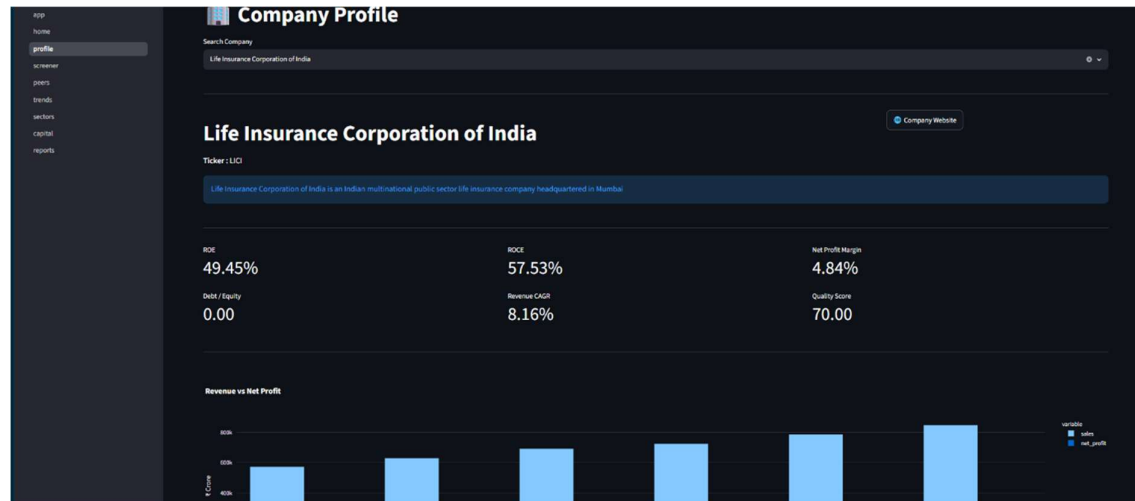
The Company Profile page provides detailed financial information for an individual company.

Users can select a company from the available list and view:

- Company overview
- Historical revenue
- Profit trends
- Balance sheet summary
- Cash flow summary
- Financial ratios

- Growth indicators

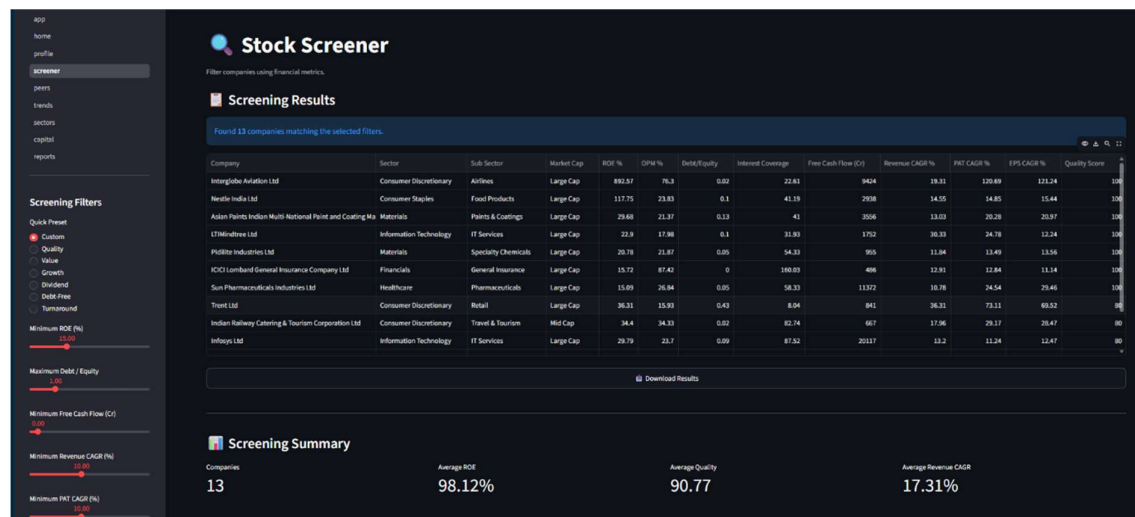
Interactive charts help users understand historical performance and identify long-term trends.



4.3 Investment Screener

The Investment Screener enables analysts to identify companies that satisfy predefined investment strategies or custom financial criteria. Instead of manually reviewing every company, users can apply filters to narrow down the investment universe based on financial performance and business quality.

This feature is particularly useful for investors looking to identify companies that match a specific investment philosophy, such as high-growth businesses, value investments, or financially stable companies.



Using the Investment Screener

Follow the steps below to use the screener.

Step 1: Open the Screener

From the navigation panel on the left side of the dashboard, click **Investment Screener**.

The screener page displays all available screening options.

Step 2: Choose a Screening Strategy

The platform provides multiple predefined investment strategies.

Examples include:

- Quality Compounder
- Value Pick
- Growth Accelerator
- Dividend Champion
- Debt-Free Blue Chip
- Turnaround Watch

These presets automatically configure financial filters based on commonly used investment methodologies.

Step 3: Apply Custom Filters

Users may also create their own screening criteria.

Typical filters include:

- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Revenue CAGR
- Profit CAGR
- Debt-to-Equity Ratio
- Current Ratio
- Market Capitalisation
- Operating Profit Margin
- Net Profit Margin
- Dividend Yield

For example, an analyst searching for financially strong companies may specify:

- ROE > 20%
- Debt-to-Equity < 0.50
- Revenue CAGR > 15%
- Positive Free Cash Flow

The platform automatically filters companies matching these conditions.

Step 4: Review Results

After applying the desired filters, click **Run Screener**.

The dashboard displays a table containing all matching companies.

Each row generally includes:

- Company Name
- Sector
- Composite Score
- ROE
- ROCE
- Revenue CAGR
- Debt-to-Equity
- Free Cash Flow
- Overall Ranking

Users can sort the results by any available metric to prioritise companies based on their investment objectives.

Step 5: Export Results

The screener allows users to export the filtered company list for further analysis.

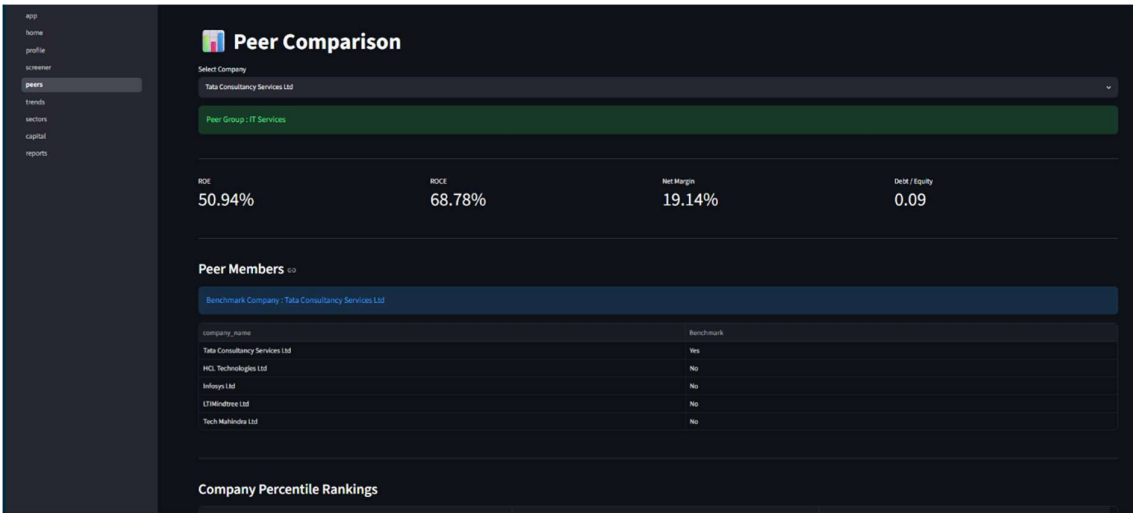
Click the **Export** button to generate an Excel file containing all selected companies and their financial metrics.

This report can be shared with other analysts or used for additional research.

4.4 Peer Analytics

The Peer Analytics page compares a selected company against other companies operating within the same industry or peer group.

Rather than evaluating a company in isolation, peer comparison provides context by showing how its financial performance ranks against competitors.



Selecting a Company

Choose a company from the available dropdown list.

The platform automatically retrieves its peer group and computes comparative financial metrics.

Peer Comparison Metrics

The comparison includes several important indicators, such as:

- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Net Profit Margin
- Revenue CAGR
- Profit CAGR
- Debt-to-Equity Ratio
- Free Cash Flow
- Composite Score

Each metric is compared with peer averages and percentile rankings.

Radar Chart

A radar chart visually compares the selected company with its peer group across multiple financial dimensions.

The chart allows analysts to quickly identify:

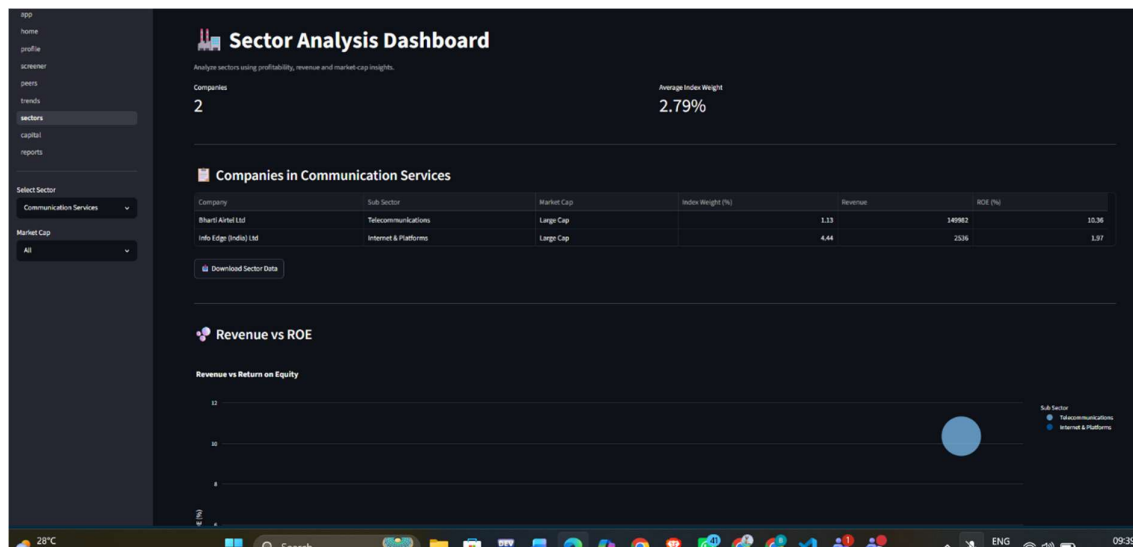
- Areas where the company outperforms competitors
- Metrics requiring improvement
- Overall financial balance

Radar charts provide an intuitive overview of relative business performance.

4.5 Sector Analytics

The Sector Analytics page provides a high-level overview of companies operating within the same business sector.

This helps analysts understand industry trends and identify sector leaders.



The page typically includes:

- Sector rankings
- Median financial ratios
- Growth comparisons
- Profitability comparisons
- Company rankings

Analysts can compare companies within the same sector instead of comparing unrelated industries.

4.6 Cash Flow Intelligence

The Cash Flow Intelligence page focuses on analysing how companies generate and utilise cash.

Unlike traditional financial ratios, this module evaluates the quality of cash generation and long-term financial sustainability.

Key indicators displayed include:

CFO Quality

Measures the relationship between operating cash flow and reported profits.

Possible classifications include:

- High Quality
- Moderate
- Accrual Risk

Capital Expenditure Intensity

Indicates how aggressively a company reinvests in business assets.

Categories include:

- Asset Light

- Moderate Investment
- Capital Intensive

Distress Indicators

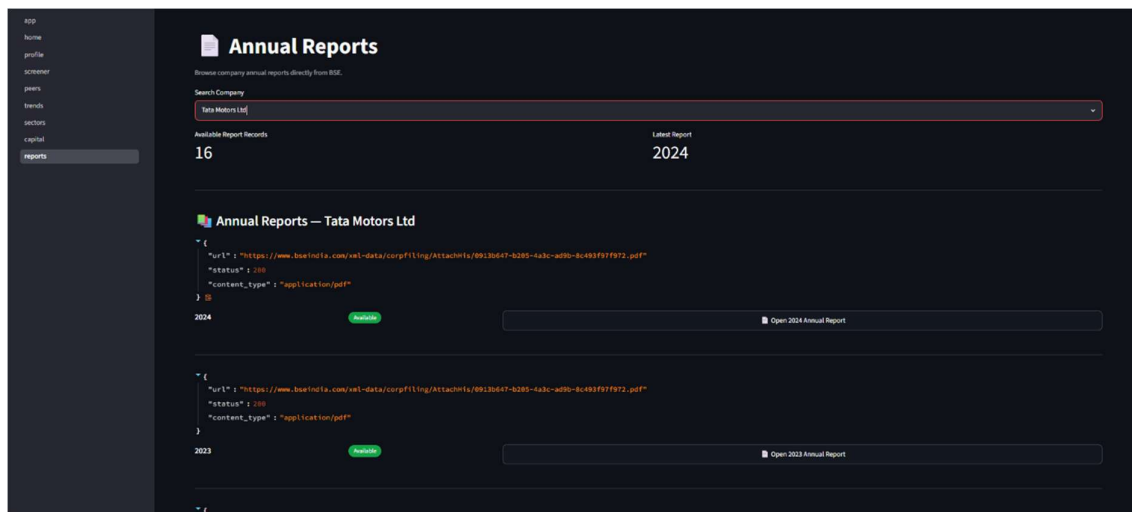
The dashboard identifies companies showing signs of financial stress, including:

- Negative operating cash flow
- High financing dependence
- Weak internal cash generation

These indicators help analysts identify potential investment risks.

4.7 Reports

The Reports page provides access to all generated analytical reports.



Users can download:

- Company Tearsheets
- Sector Reports
- Portfolio Summary
- Screener Results
- Peer Comparison Reports

Reports are generated in PDF or Excel format depending on the report type.

5. Generating PDF Tearsheets

The Financial Intelligence Platform automatically generates professional PDF tearsheets for individual companies.

A tearsheet summarises the financial performance of a company in a concise, visually rich report suitable for investment research and presentations.

Generating Tearsheets

Run the following command:

```
python src/reports/batch_tearsheets.py
```

The reporting engine automatically processes all available companies and generates individual PDF files.

Generated reports are stored in the reports directory.

Example:

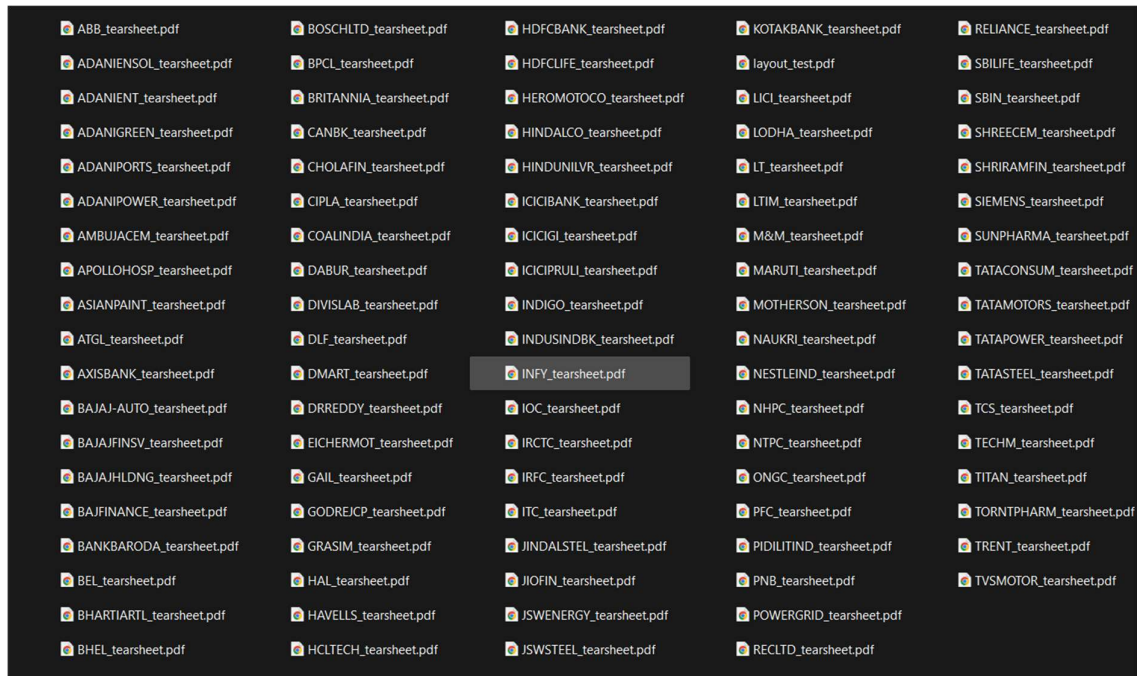
reports/

Reliance.pdf

Infosys.pdf

TCS.pdf

HDFC_Bank.pdf



Contents of a Company Tearsheet

Each tearsheet typically contains:

- Company profile
- Revenue trend
- Profit trend
- Key financial ratios
- Cash flow summary
- Capital allocation analysis
- Radar chart

- AI-generated pros and cons
- Overall financial summary

These reports provide analysts with a quick yet comprehensive overview of company performance without requiring manual calculations.

6. Using the REST API

The Financial Intelligence Platform provides a RESTful API built with **FastAPI**, allowing external applications and developers to access financial data programmatically.

The API exposes company information, financial metrics, peer comparisons, screening results, and other analytical outputs through HTTP endpoints. It can be integrated into web applications, mobile applications, or third-party analytics tools.

6.1 Starting the API Server

Before making API requests, start the FastAPI server.

```
uvicorn src.api.main:app --reload
```

If the server starts successfully, you should see output similar to:

```
INFO: Uvicorn running on http://127.0.0.1:8000
```

The API is now ready to accept requests.

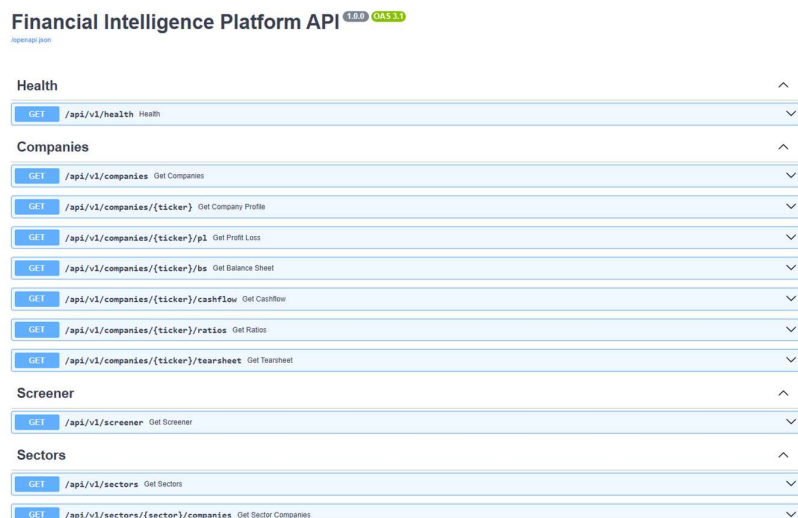
6.2 Interactive API Documentation

FastAPI automatically generates interactive API documentation.

Open your browser and navigate to:

```
http://127.0.0.1:8000/docs
```

The Swagger UI displays all available endpoints and allows users to test requests directly from the browser without using external tools.



6.3 Calling API Endpoints Using curl

The following examples demonstrate how to interact with the API using the curl command-line utility.

Note: Replace the example endpoint paths below with the actual endpoints implemented in your project if they differ.

Example 1 – Health Check

This endpoint verifies that the API server is running correctly.

```
curl -X GET "http://127.0.0.1:8000/health"
```

Expected Response

```
{  
  "status": "healthy"  
}
```

Example 2 – Get Company Information

Retrieve information for a specific company.

```
curl -X GET "http://127.0.0.1:8000/company/TCS"
```

Example Response

```
{  
  "company": "TCS",  
  "sector": "Information Technology",  
  "market_cap": "...",  
  "roe": "...",  
  "roce": "..."  
}
```

Example 3 – Investment Screener

Retrieve companies satisfying predefined screening criteria.

```
curl -X GET "http://127.0.0.1:8000/screener"
```

Example Response

```
[  
  {  
    "company": "ABC Ltd",  
    "score": 92.4  
  },  
  {  
    "company": "XYZ Ltd",  
    "score": 89.7  
  }  
]
```

```
}  
]
```

Example 4 – Sector Analytics

Retrieve companies belonging to a specific sector.

```
curl -X GET "http://127.0.0.1:8000/sector/IT"
```

Example Response

```
[  
  {  
    "company": "TCS"  
  },  
  {  
    "company": "Infosys"  
  }  
]
```

Example 5 – Financial Ratios

Retrieve calculated financial ratios.

```
curl -X GET "http://127.0.0.1:8000/ratios/TCS"
```

Example Response

```
{  
  "roe": 23.5,  
  "roce": 31.7,  
  "debt_equity": 0.08  
}
```

6.4 Common API Response Codes

HTTP Status Code Meaning

200	Request completed successfully
400	Invalid request parameters
404	Requested resource not found
422	Validation error
500	Internal server error

These status codes help users identify whether a request was processed successfully or if corrective action is required.

7. Troubleshooting

This section describes common issues that users may encounter while using the Financial Intelligence Platform and suggests possible solutions.

Issue 1 – Dashboard Does Not Open

Possible Cause

The Streamlit server is not running.

Solution

Verify that the following command has been executed successfully.

```
streamlit run src/dashboard/app.py
```

Ensure that no firewall or port conflict is preventing Streamlit from starting.

Issue 2 – REST API Is Not Accessible

Possible Cause

The FastAPI server has not been started or is running on a different port.

Solution

Start the API using:

```
uvicorn src.api.main:app --reload
```

Then access:

```
http://127.0.0.1:8000/docs
```

Issue 3 – Database File Not Found

Possible Cause

The SQLite database has not been created.

Solution

Execute the ETL pipeline.

```
python src/etl/load_database.py
```

This creates and populates the database with validated financial data.

Issue 4 – No Companies Displayed in Dashboard

Possible Cause

Financial data has not been loaded or analytical metrics have not been generated.

Solution

Run both commands in sequence.

```
python src/etl/load_database.py
```

Followed by:


```
python src/analytics/populate_financial_ratios.py
```

Restart the dashboard after the analytics process completes.

Issue 5 – PDF Reports Are Not Generated

Possible Cause

The reporting script was not executed or required directories do not exist.

Solution

Run the report generation script.

```
python src/reports/batch_tearsheets.py
```

Verify that the output directory exists and has write permissions.

Issue 6 – ModuleNotFoundError

Possible Cause

Project dependencies have not been installed or the virtual environment is inactive.

Solution

Activate the virtual environment and reinstall the dependencies.

```
pip install -r requirements.txt
```

Issue 7 – Import Errors

Possible Cause

Incorrect project directory or missing Python packages.

Solution

Ensure that commands are executed from the project root directory and verify that all required packages have been installed successfully.

Issue 8 – Empty Screener Results

Possible Cause

The selected filters are too restrictive.

Solution

Relax one or more filter conditions and rerun the screener.

8. Best Practices

To obtain reliable analytical results, users should follow these recommended practices.

- Always execute the ETL pipeline before running analytical modules.
- Refresh the SQLite database whenever the source Excel datasets are updated.
- Compare companies within the same sector for meaningful benchmarking.

- Use multiple financial indicators rather than relying on a single ratio.
- Review both quantitative metrics and qualitative insights before making investment decisions.
- Regularly generate updated reports to reflect the latest available financial data.
- Keep project dependencies updated using the latest compatible versions.

Following these practices helps ensure accurate analysis and improves the reliability of generated insights.

9. Conclusion

The Financial Intelligence Platform provides a comprehensive environment for financial analysis by integrating data engineering, financial analytics, reporting, and interactive visualisation into a single workflow.

Through the Streamlit dashboard, analysts can explore company performance, compare peers, evaluate sectors, and identify investment opportunities using predefined or customised screening strategies. The reporting engine simplifies communication through professionally formatted PDF tearsheets, while the REST API enables seamless integration with external applications.

By following the procedures described in this guide, users can efficiently operate the platform, generate analytical reports, troubleshoot common issues, and make effective use of the available financial insights.

Appendix

Useful Commands

Load Data

```
python src/etl/load_database.py
```

Generate Analytics

```
python src/analytics/populate_financial_ratios.py
```

Generate PDF Reports

```
python src/reports/batch_tearsheets.py
```

Launch Dashboard

```
streamlit run src/dashboard/app.py
```

Launch API

```
uvicorn src.api.main:app --reload
```

Run Tests

```
pytest tests -v
```